COMP9032 Lab 2

Sept. 2026

1. Objectives

In this lab, you will learn AVR programming on

- function
- parallel I/O

2. Preparation

For this lab, you will use the lab board. It is required that you:

- read Introduction to AVR Microprocessor Development Board on how to use the board, and
- download and install the following Arduino software that is available on the course website.
    - arduino-1.0.6-windows.exe

3. Tasks

There are two tasks in this lab.

Task 1: (10 marks, **due in your Week 4 lab session**)

Test the lab boards distributed to you. It is required that

- Before the test, read all documents about the lab board, which are available on the Labs page of the course website
- Test the board by following the test procedure provided in the Lab Board Information.zip file
    - There may already be some wire connections on the board. But you still need to identify the ports and check each connection as specified in the test sheet.
    - After the test, pull off all wires and make the board ready for the coming lab exercises.
- Upload the test report to your group channel in MS Teams before the due time. **Notify your tutor if there are any problems with your lab board**.

  Your test report must not be modified after your lab session in Week 4. Any changes made afterward will be considered late submission and no marks will be awarded.

Task 2 (90 marks, **due in your Week 5 lab session**)

1) Write an assembly program that consists of a function. The function reads a text from the program memory and returns the total number of words in the text.

In the main program, the function is called for a set of different texts under the control of a push button. Pressing the button, the function is performed on one text, and the resulting word count is displayed on the LED bar. Pressing the button again, the function is performed for another text in the program memory.

It is assumed that:

- Each text is terminated by ETX (End of Text, 0x03).
- Words are separated by a single space character (0x20).
- Punctuation marks are considered as part of the words. For example, “Hello, everyone!” contains two words.

It is also assumed that the maximum word count is limited by 4 bits. When the text length is larger than the limit, an exception happens, which will trigger the LED bar flashes until another push button is pressed.

The LED bar on the lab board is divided into two parts: the upper 4 LEDs for displaying word count and the lower 6 LEDs for indicating exception.

2) Create a demo video to verify your design for the following five text cases
    - Case 1: a small text that contains a short sentence of less than 15 words
    - Case 2: a small text that contains two short sentences with a total length of less than 15 words
    - Case 3: a text that has exactly 15 words
    - Case 4: a text that has exactly 16 words
    - Case 5: a long text that has more than 16 words.

    You need to design how to place the five texts in the memory for easy access during the program execution.

    For the video demo, adding annotations – either verbal or text—will make your presentation more effective. It is up to you to decide what should be included in your demo.

4. Peer Assessment

Task 2 will be assessed by your peers and tutors. The assessment is based on

1) Explanation of design and implementation (25 marks)
    a. Function (5 marks)
    b. Use of push button for activating function call (5 marks)
    c. Use of push button for stopping LED flashing (5 marks)
    d. Texts in the program memory (5 marks)
    e. Overall code structure (5 marks)
2) Quality of the design and implementation (25 marks)
    a. Function (5 marks)
    b. Use of push button for activating function call (5 marks)
    c. Use of push button for stopping LED flashing (5 marks)
    d. Texts in the program memory (5 marks)
    e. Overall code structure (5 marks)
3) Video demonstration (8 marks)
4) Five tests (32 marks)
    a. Case 1: The word count is correctly displayed on LEDs. (5 marks)

b. Case 2: The word count is correctly displayed on LEDs. (5 marks)
c. Case 3: The word count is correctly displayed on LEDs (5 marks)
d. Case 4: Exception is triggered, and LED flash can be stopped by the push button. (9 marks)
e. Case 5: Exception is triggered, and LED flash can be stopped by the push button. (9 marks)